

Find and Fix: Illegal Cancellation and Lost Solutions

Answer Key

Precalculus

Quick Answers

- | | |
|---|--|
| 1. (a) all solutions = 0, 60, 180, 300 degrees, — | 6. (a) correct simplification = $1 + 2 \cos(\theta)$, (b) |
| 2. (a) all solutions = 30, 150, 210, 330 degrees, — | value at 60 degrees = 2, — |
| 3. (a) all solutions = 60, 90, 270, 300 degrees, — | 7. (a) all solutions = 0, 120, 240 degrees, — |
| 4. (a) all solutions = 0, 270 degrees, — | 8. (a) all solutions = 0, 30, 150, 180, 210, 330 |
| 5. (a) all solutions = 15, 75, 195, 255 degrees, — | degrees, — |
-

What is verified

9 of 17 answers machine-checked against SymPy, across 8 problems.
 — marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Precalculus

HSF-TF.C.8 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

1. If they got 60, 300: divided both sides by sine, which throws away every angle where the sine is zero
2. If they got 30, 150: took only the positive square root, dropping the whole negative case
3. If they got 60, 300: cancelled a cosine from both sides, losing the two angles at which the cosine is zero
4. If they got 0, 180, 270: squared both sides and reported every root of the squared equation without testing them in the original
5. If they got 15, 75: solved the doubled angle over the original interval instead of the doubled one, so half the solutions were never found
6. If they got 3: cancelled the sine out of the double-angle term as well and lost the cosine factor with it
7. If they got 120, 240: decided the factor giving a cosine of one had no solution and dropped it
8. If they got 30, 150: cancelled the sine factor and also solved the doubled angle over the original interval, losing solutions twice over

1. $2 \sin \theta \cos \theta = \sin \theta$ on $0^\circ \leq \theta < 360^\circ$. Student divided by $\sin \theta$ and got $60^\circ, 300^\circ$.

(a) Move everything to one side and factor — never divide by something that can be zero.

$$2 \sin \theta \cos \theta - \sin \theta = 0 \implies \sin \theta (2 \cos \theta - 1) = 0$$

$\sin \theta = 0$ gives $\theta = 0^\circ, 180^\circ$; $\cos \theta = \frac{1}{2}$ gives $\theta = 60^\circ, 300^\circ$.

$$\theta = 0^\circ, 60^\circ, 180^\circ, 300^\circ$$

(b) *Instructor-judged.* The response must say that dividing by $\sin \theta$ is legal only where $\sin \theta \neq 0$, and that the angles where it *is* zero are precisely the two solutions destroyed. Full credit also states the repair: collect on one side and factor, so the sine survives as a factor. Noting only that two answers are missing is partial credit.

2. $4 \sin^2 \theta - 1 = 0$ on $0^\circ \leq \theta < 360^\circ$. Student took one square root and got $30^\circ, 150^\circ$.

(a) Factor as a difference of squares, which cannot lose a branch:

$$(2 \sin \theta - 1)(2 \sin \theta + 1) = 0 \implies \sin \theta = \frac{1}{2} \text{ or } \sin \theta = -\frac{1}{2}.$$

The positive case gives $30^\circ, 150^\circ$; the negative case gives $210^\circ, 330^\circ$.

$$\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

(b) *Instructor-judged.* The response must say a square equal to a positive number has *two* square roots, so $\sin \theta = \pm \frac{1}{2}$, and only the positive branch was kept. Each branch supplies two angles per turn, so four solutions exist. Recommending the difference-of-squares factorisation earns full credit.

3. $2 \cos^2 \theta = \cos \theta$ on $0^\circ \leq \theta < 360^\circ$. Student divided by $\cos \theta$ and got $60^\circ, 300^\circ$.

(a)

$$2 \cos^2 \theta - \cos \theta = 0 \implies \cos \theta (2 \cos \theta - 1) = 0$$

$\cos \theta = 0$ gives $\theta = 90^\circ, 270^\circ$; $\cos \theta = \frac{1}{2}$ gives $\theta = 60^\circ, 300^\circ$.

$$\theta = 60^\circ, 90^\circ, 270^\circ, 300^\circ$$

(b) *Instructor-judged.* The response must identify the cancelled cosine and say that cancelling silently assumes $\cos \theta \neq 0$ — an assumption the problem never grants. Factoring keeps both cases. Full credit requires naming the assumption, not just listing the two missing right angles.

4. $\sin \theta + 1 = \cos \theta$ on $0^\circ \leq \theta < 360^\circ$. Student squared and reported $0^\circ, 180^\circ, 270^\circ$.

(a) The squared equation does give those three candidates, but squaring is not reversible, so each must be tested in the *original* equation:

$\theta = 0^\circ : 0 + 1 = 1 \text{ and } \cos 0^\circ = 1$	✓
$\theta = 180^\circ : 0 + 1 = 1 \text{ but } \cos 180^\circ = -1$	×
$\theta = 270^\circ : -1 + 1 = 0 \text{ and } \cos 270^\circ = 0$	✓

$$\theta = 0^\circ, 270^\circ$$

(b) *Instructor-judged.* The response must say squaring can create roots the original never had, so every candidate has to be substituted back, and must carry out that test on 180° to show the two sides disagree. Saying only that one answer is wrong, without the reason squaring produces it, is partial credit.

5. $2 \sin 2\theta = 1$ on $0^\circ \leq \theta < 360^\circ$. Student reported $15^\circ, 75^\circ$.

(a) Put $u = 2\theta$. As θ runs over $[0^\circ, 360^\circ)$, u runs over $[0^\circ, 720^\circ)$:

$$\sin u = \frac{1}{2} \implies u = 30^\circ, 150^\circ, 390^\circ, 510^\circ.$$

Halving each: $\theta = 15^\circ, 75^\circ, 195^\circ, 255^\circ$.

$$\theta = 15^\circ, 75^\circ, 195^\circ, 255^\circ$$

(b) *Instructor-judged.* The response must say the doubled angle sweeps twice as far, so its interval ends at 720° , and stopping at 360° discards the second pass. Full credit states the repair: substitute for the inside angle, multiply both endpoints by the coefficient, solve, then divide every answer back.

6. Simplify $\frac{\sin \theta + \sin 2\theta}{\sin \theta}$. Student cancelled every $\sin \theta$ and wrote 3.

(a) Expand the double angle first, then factor the numerator before dividing:

$$\frac{\sin \theta + 2 \sin \theta \cos \theta}{\sin \theta} = \frac{\sin \theta (1 + 2 \cos \theta)}{\sin \theta} = 1 + 2 \cos \theta \quad (\sin \theta \neq 0).$$

$$1 + 2 \cos \theta$$

(b) At $\theta = \frac{\pi}{3}$, $\cos \theta = \frac{1}{2}$, so the expression is $1 + 2 \left(\frac{1}{2}\right)$.

$$2$$

(c) *Instructor-judged.* The response must say a common factor may only be cancelled from a whole term: the second term is $2 \sin \theta \cos \theta$, so removing its $\sin \theta$ leaves $2 \cos \theta$, not a bare 2. Full credit requires naming the dropped cosine factor; a correct answer shows the numerator written as $\sin \theta(1 + 2 \cos \theta)$ before dividing.

7. $\cos 2\theta = \cos \theta$ on $0^\circ \leq \theta < 360^\circ$. Student reported $120^\circ, 240^\circ$ and discarded $\cos \theta = 1$.

(a) The factoring is correct: $(2 \cos \theta + 1)(\cos \theta - 1) = 0$. Both factors matter.

$$\cos \theta = -\frac{1}{2} \implies \theta = 120^\circ, 240^\circ; \quad \cos \theta = 1 \implies \theta = 0^\circ.$$

$$\theta = 0^\circ, 120^\circ, 240^\circ$$

(b) *Instructor-judged.* The response must say that $\cos \theta = 1$ is perfectly attainable — it happens at 0° , which the half-open interval includes — so that factor contributes a solution. Full credit notes that 360° is excluded, which is why the angle appears once rather than twice. Supplying the missing angle without saying why the student wrongly rejected it is partial credit.

8. $2 \sin \theta \cos 2\theta = \sin \theta$ on $0^\circ \leq \theta < 360^\circ$. Student cancelled $\sin \theta$, used $2\theta \in [0^\circ, 360^\circ)$, and reported $30^\circ, 150^\circ$.

(a) Factor rather than cancel, then widen the interval for the doubled angle:

$$\sin \theta (2 \cos 2\theta - 1) = 0$$

$\sin \theta = 0$ gives $\theta = 0^\circ, 180^\circ$. For the other factor, $u = 2\theta$ runs over $[0^\circ, 720^\circ)$ and $\cos u = \frac{1}{2}$ at $u = 60^\circ, 300^\circ, 420^\circ, 660^\circ$, so $\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$.

$\theta = 0^\circ, 30^\circ, 150^\circ, 180^\circ, 210^\circ, 330^\circ$
--

(b) *Instructor-judged.* Two independent faults must be named, with the losses attributed. Dividing by $\sin \theta$ cost 0° and 180° . Solving the doubled angle over the un-widened interval cost 210° and 330° . Naming one fault only is half credit; a correct response says which missing solutions came from which fault.